

Name: Malaika Asim

Sap ID: 74507

Semester: 3rd

Department: BSCY

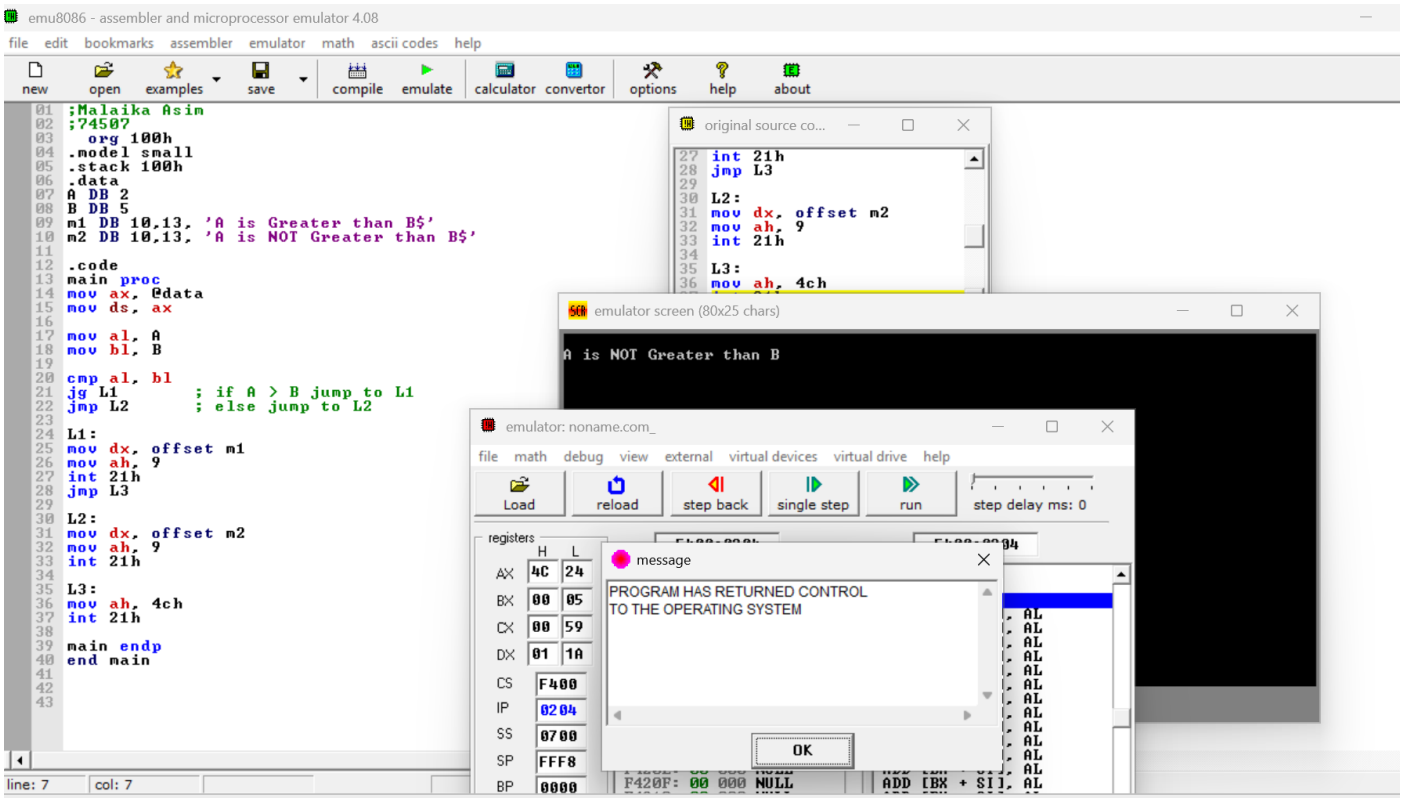
Course: COAL Lab

Instructor: Ma’am Rehana Sanam

LAB 09

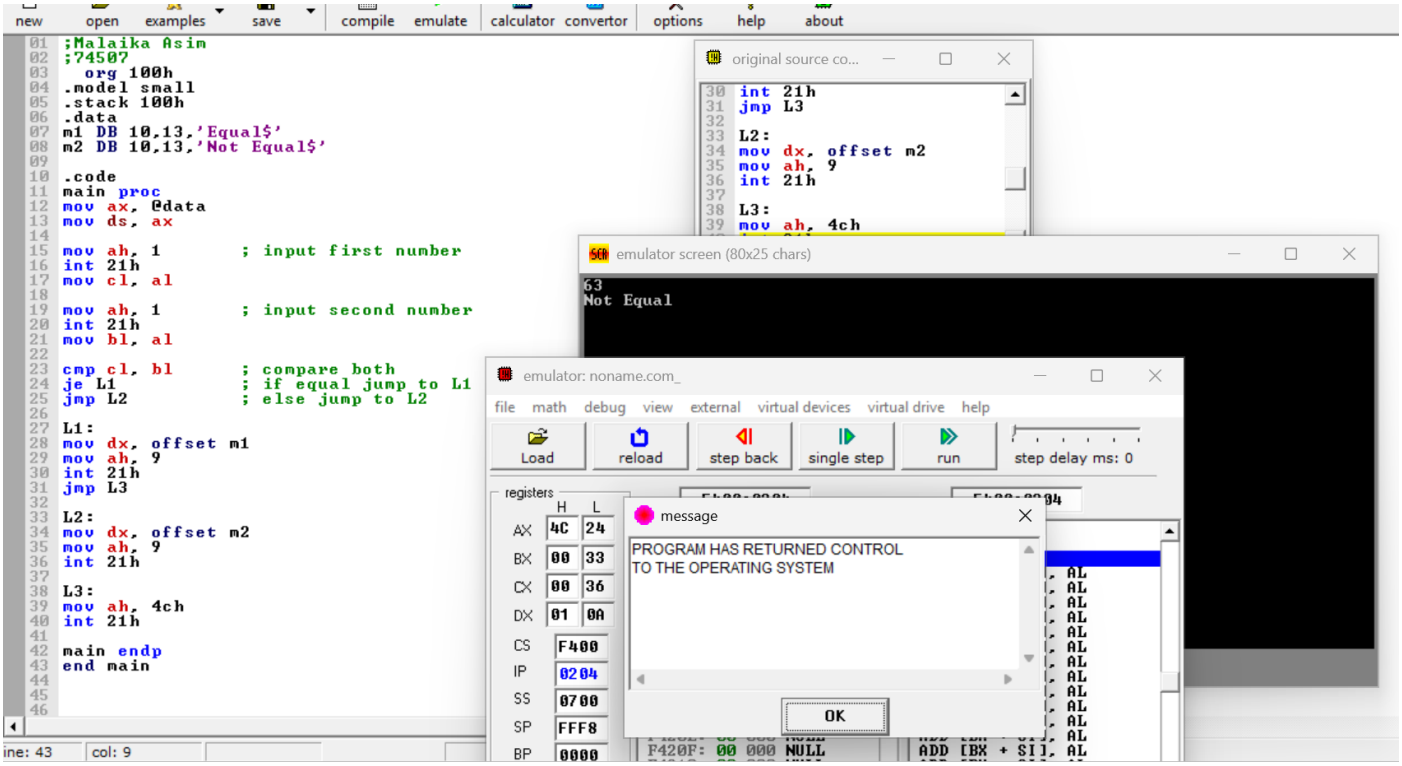
Task 01

Write an assembly language that compares the values A and B and prints a message if the value in A is greater than B or not.



Task 02

Write an assembly language that compares two numbers and prints a message if numbers are equal or not.



```
;Malaika Asim
;74507
  org 100h
.model small
.stack 100h
.data
m1 DB 10,13,'Equal$'
m2 DB 10,13,'Not Equal$'

.code
main proc
mov ax, @data
mov ds, ax

mov ah, 1        ; input first number
int 21h
mov cl, al

mov ah, 1        ; input second number
int 21h
mov bl, al

cmp cl, bl       ; compare both
je L1            ; if equal jump to L1
jmp L2           ; else jump to L2

L1:
mov dx, offset m1
mov ah, 9
int 21h
jmp L3

L2:
mov dx, offset m2
mov ah, 9
int 21h

L3:
mov ah, 4ch
int 21h

main endp
end main
```

```
int 21h
jmp L3

L2:
mov dx, offset m2
mov ah, 9
int 21h

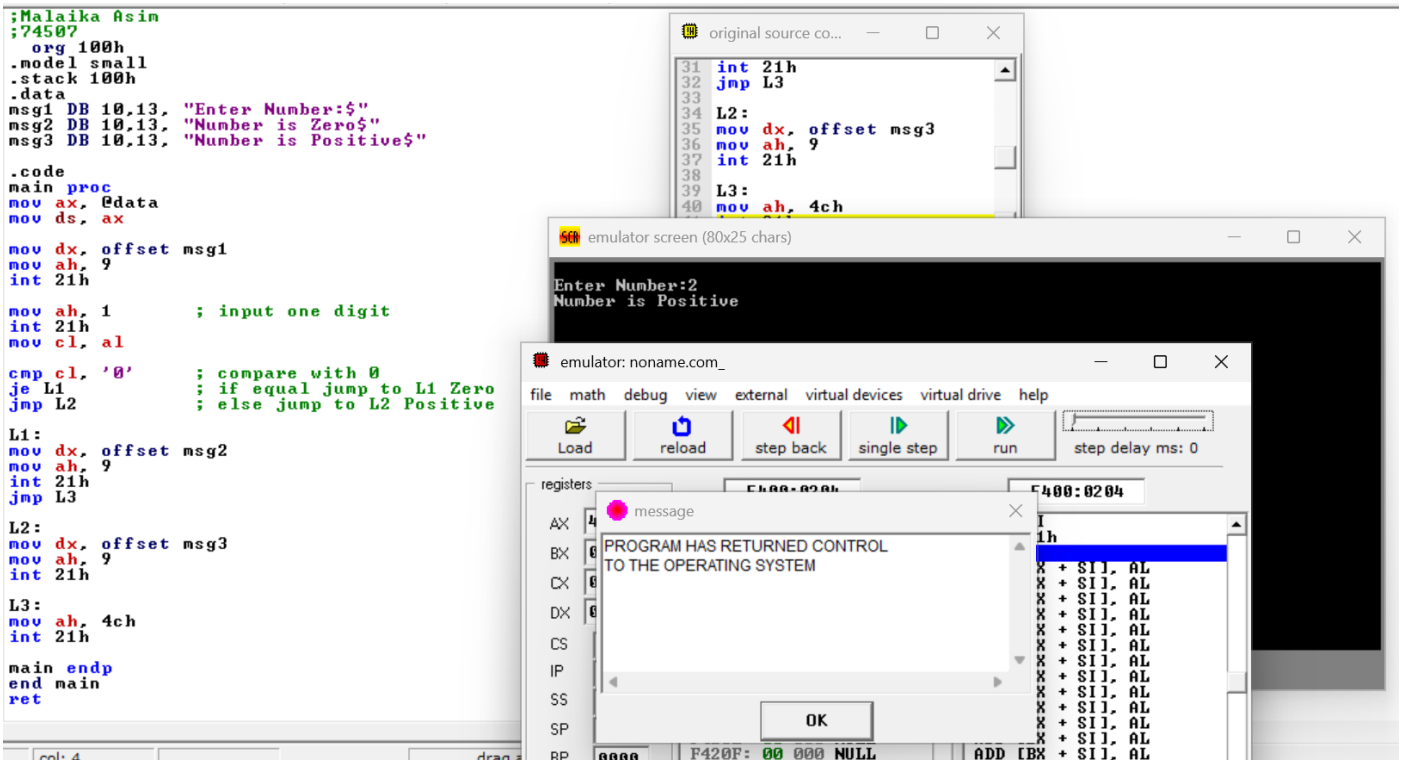
L3:
mov ah, 4ch
```

```
63
Not Equal
```

```
PROGRAM HAS RETURNED CONTROL
TO THE OPERATING SYSTEM
```

Task 03

Write an assembly language program that allow user to input one-digit number and determine the number is positive or zero.



```
;Malaika Asim
;74507
  org 100h
.model small
.stack 100h
.data
msg1 DB 10,13, "Enter Number:$"
msg2 DB 10,13, "Number is Zero$"
msg3 DB 10,13, "Number is Positive$"

.code
main proc
mov ax, @data
mov ds, ax

mov dx, offset msg1
mov ah, 9
int 21h

mov ah, 1        ; input one digit
int 21h
mov cl, al

cmp cl, '0'      ; compare with 0
je L1            ; if equal jump to L1 Zero
jmp L2           ; else jump to L2 Positive

L1:
mov dx, offset msg2
mov ah, 9
int 21h
jmp L3

L2:
mov dx, offset msg3
mov ah, 9
int 21h

L3:
mov ah, 4ch
int 21h

main endp
end main
ret
```

```
int 21h
jmp L3

L2:
mov dx, offset msg3
mov ah, 9
int 21h

L3:
mov ah, 4ch
```

```
Enter Number:2
Number is Positive
```

```
PROGRAM HAS RETURNED CONTROL
TO THE OPERATING SYSTEM
```

Task 04

What is the difference between an unconditional jump and a conditional jump in detail with examples?

Feature	Unconditional Jump	Conditional Jump
Meaning	Always jumps, no checking	Jumps only if condition is true
Syntax	JMP targetLabel	Jcond destination Ex: JE, JNE, JG, JL, JGE
Depends on Flags?	No	Yes - checks ZF, CF, SF, etc after CMP
Comparison Needed?	No	Yes - normally used after CMP
Example Code	L1: mov dl,'b' / mov ah,2 / int 21h / JMP L1 - infinite loop printing b	cmp dl, cl / je label1 - if equal jump to label1 else continue